LAB REPORT

DAFFODIL INTERNATIONAL UNIVERSITY
Department Of
Computing & Information System
Faculty Of Science & Information Technology

Instructor:
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LAB REPORT- 03

Course Title : Structured Programming
Course Code : CIS 115
Subject : Switch , Loop

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Switch

Problem No: 6.1

Problem Name: Write a program that read a digit and display by spelling.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("enter a number:");
    scanf("%d",&n);
    switch(n){
        case 0:
            printf("Zero");
            break;
        case 1:
            printf("One");
            break;
        case 2:
            printf("Two");
            break;
        case 3:
            printf("Three");
            break;
        case 4:
            printf("Four");
            break;
```

```
case 5:
            printf("Five");
            break;
        case 6:
            printf("Six");
            break;
        case 7:
            printf("Seven");
            break;
        case 8:
            printf("Eight");
            break;
        case 9:
            printf("Nine");
            break;
        default:
            printf("Not a sigle digit 
number");
    }
    return 0;
}
```

Output:

```
PS H:\Programming\Coding Practice\Basic C\exerise class work\6> cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\6\" ; if ($?) { gcc code1.c -o cod
e1 } ; if ($?) { .\code1 }
enter a number:7
Seven
PS H:\Programming\Coding Practice\Basic C\exerise class work\6>
```

Problem No: 6.2

Problem Name: Write a program that read any number and display equivalent roman number.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("enter a number:");
    scanf("%d",&n);
    printf("Equivalent roman number is:");
    int h=n/1000;
    for(int i=1;i<h;i++){
        printf("M");
    }
    int s=(n%1000)/100;
    switch(s){
        case 1:
        printf("C");
        break;
        case 2:
        printf("CC");
        break;
        case 3:
        printf("CCC");
        break;
        case 4:
        printf("CD");
        break;
```

```
        case 5:
        printf("D");
        break;
        case 6:
        printf("DC");
        break;
        case 7:
        printf("DCC");
        break;
        case 8:
        printf("DCCC");
        break;
        case 9:
        printf("CM");
        break;
    }
    int d=(n%100)/10;
    switch(d){
        case 1:
        printf("X");
        break;
        case 2:
        printf("XX");
        break;
        case 3:
        printf("XXX");
        break;
        case 4:
        printf("XL");
        break;
        case 5:
        printf("L");
        break;
```

```
        case 6:
        printf("LX");
        break;
        case 7:
        printf("LXX");
        break;
        case 8:
        printf("LXXX");
        break;
        case 9:
        printf("XC");
        break;
    }
    int a=n%10;
     switch(a){
        case 1:
        printf("I");
        break;
        case 2:
        printf("II");
        break;
        case 3:
        printf("III");
        break;
        case 4:
        printf("IV");
        break;
        case 5:
        printf("V");
        break;
        case 6:
        printf("VI");
        break;
```

```
        case 7:
        printf("VII");
        break;
        case 8:
        printf("VIII");
        break;
        case 9:
        printf("IX");
        break;
    }
    return 0;
}
```

Output:

```
                                                   > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\6\" ; if ($?) { gcc code2.c -o cod
e2 } ; if ($?) { .\code2 }
enter a number:12
Equivalent roman number is:XII
PS H:\Programming\Coding Practice\Basic C\exerise class work\6>
```

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Loop

Problem No: 9.1

Problem Name: Write a program that read a positive integer and display its factorial..

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a positive integer:");
    scanf("%d",&n);
    int product=1;
    for(int i=1;i<=n;i++){
        product=product*i;
    }
    printf("the factorial is
%d",product);
    return 0;
}
```

Output:

```
                                              > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code1.c -o cod
e1 } ; if ($?) { .\code1 }
Enter a positive integer:5
the factorial is 120
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

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Problem No: 9.2

Problem Name: Write a program that read any positive integer and display sum of its digit.

Source Code:

```c
#include<stdio.h>
int main(){
    int n;
    printf("Enter a number :");
    scanf("%d",&n);
    int last_digit=0;
    int result=0;
    while(n!=0){
        last_digit=n%10;
        result=result+last_digit;
        n=n/10;
    }
    printf("sum of digits is %d",result);
    return 0;
}
```

Output:

```
> cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc tempCodeRunner
File.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter a number :123
sum of digits is 6
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.3

Problem Name: Write a program that read any positive integer and display reverse.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter any integer: ");
    scanf("%d",&n);
    int ld=0;
    int reverse=0;
    while(n!=0){
        ld=n%10;
        reverse=reverse*10;
        reverse=reverse+ld;
        n=n/10;
    }
    printf("the reverse number
is:%d",reverse);
    return 0;
}
```

Output:

```
                                                  > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code3.c -o cod
e3 } ; if ($?) { .\code3 }
Enter any integer: 12345
the reverse number is:54321
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
                                              Ln 16, Col 2    Spaces: 4    UT
```

Problem No: 9.4

Problem Name: Write a program that read any decimal number and display equivalent binary number.

Source Code:

```
#include<stdio.h>
int main(){
    int a[10],n,i,j;
    printf("Enter any decimal number: ");
    scanf("%d",&n);
    for(i=0;n>0;i++){
        a[i]=n%2;
        n=n/2;
    }
    for(j=i-1;j>=0;j--){
        printf("%d",a[j]);
    }
    return 0;
}
```

Output:

```
> cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code4.c -o cod
e4 } ; if ($?) { .\code4 }
Enter any decimal number: 50
110010
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
Ln 14, Col 2   Spaces: 4   UTF
```

Problem No: 9.5

Problem Name: Write a program that read any decimal number and display equivalent octal number.

Source Code:

```
#include<stdio.h>
int main(){
    int a[32],n,i=0,j;
    printf("Enter a number :");
    scanf("%d",&n);
    if(n==0){
        printf("octal: 0 \n");
        return 0;
    }
    while(n>0){
        a[i]=n%8;
        n=n/8;
        i++;
    }
    printf("octal: ");
    for(j=i-1;j>=0;j--){
        printf("%d",a[j]);
    }
    return 0;
}
```

Output:

```
> cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code5.c -o cod
e5 } ; if ($?) { .\code5 }
Enter a number :11
octal: 13
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

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Problem No: 9.6

Problem Name: Write a program that read any decimal number and display equivalent hexadecimal number.

Source Code:

```
#include<stdio.h>
int main(){
    int a[100],n,i=0,j;
    printf("Enter a number :");
    scanf("%d",&n);
    if(n==0){
        printf("Hexa-decimal: 0 \n");
        return 0;
    }
    while(n>0){
        int r=n%16;
        if(r<10){
            a[i]=r+'0';
        }
        else{
            a[i]=r-10+'A';
        }
        n=n/16;
        i++;
    }
    printf("Hexa-decimal: ");
    for(j=i-1;j>=0;j--){
        printf("%c",a[j]);
    }
    return 0;
}
```

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Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code6.c -o code6 } ; if ($?) { .\code6 }
Enter a number :15
Hexa-decimal: F
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.7

Problem Name: Write a program that read two numbers and display GCD(greatest common divisor).

Source Code:

```
#include<stdio.h>
int main(){
    int x,y,gcd;
    printf("Enter two number: ");
    scanf("%d %d",&x,&y);
    for(int i=1;i<=x && i<=y;i++){
        if(x%i==0 && y%i==0){
            gcd=i;
        }
    }
    printf("the greatest common divisor is %d",gcd);
    return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code7.c -o code7 } ; if ($?) { .\code7 }
Enter two number: 10
15
the greatest common divisor is 5
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.8

Problem Name: Write a program that read two numbers and display LCM(least common multiple).

Source Code:

```
#include<stdio.h>
int main(){
    int x,y,lcm,max;
    printf("Enter two number: ");
    scanf("%d %d",&x,&y);
    max=(x>y)? x:y;
    while(1){
        if(max%x==0 && max%y==0){
            lcm=max;
            break;
        }
        max++;
    }
    printf("the least common multiple is 
%d",lcm);
    return 0;
}
```

Output:

```
> cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code8.c -o cod
e8 } ; if ($?) { .\code8 }
Enter two number: 12
16
the least common multiple is 48
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.9

Problem Name: Write a program that read two numbers(x,y) and display x'(x to the power y).

Source Code:

```c
#include<stdio.h>
int main(){
    int x,y;
    printf("Enter two number: ");
    scanf("%d %d",&x,&y);
    int power=1;
    for(int i=1;i<=y;i++){
        power=power*x;
    }
    printf("%d",power);
    return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code9.c -o code9 } ; if ($?) { .\code9 }
Enter two number: 2
8
256
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.17

Problem Name: Write a program that read any integer and its digital root.

Source Code:

```c
#include<stdio.h>
int main(){
    int n,r;
    printf("Enter a number:");
    scanf("%d",&n);


    while(n>9){
        r=0;
        while(n>0){
            r=r+n%10;
            n=n/10;
        }
        n=r;
    }
    printf("%d",n);
    return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code17.c -o code17 } ; if ($?) { .\code17 }
Enter a number:16
7
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.18

Problem Name: Write a program that read any integer and test prime or not prime.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("Enter a number:");
    scanf("%d",&n);
    int isprime=1;
    for(int i=2;i<n;i++){
        if(n%i==0){
            isprime=0;
        }
    }
    if(isprime==1){
        printf("Prime");
    }
    else{
        printf("not prime");
    }
    return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code18.c -o code18 } ; if ($?) { .\code18 }
Enter a number:2
Prime
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.19

Problem Name: Write a program that read any integer and test prime or not prime.

Source Code:

```
#include<stdio.h>
int main(){
    int n,isprime;
    printf("Enter a number:");
    scanf("%d",&n);
    if(n<2){
        isprime=0;
    }
    else{
        isprime=1;
    }
    for(int i=2;i<n;i++){
        if(n%i==0){
            isprime=0;
        }
    }
    if(isprime==1){
        printf("Prime");
    }
    else{
        printf("not prime");
    }
    return 0;
}
```

Output:

```
                                                               > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code19.c -o co
de19 } ; if ($?) { .\code19 }
Enter a number:20
not prime
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.20
Problem No: 9.21
Problem No: 9.22
Problem No: 9.23

Same As Problem No-19

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Problem No: 9.24

Problem Name: Write a program that print all prime numbers from 1 to n .

Source Code:

```
#include<stdio.h>
#include<math.h>
int main(){
    int isprime,t;
    for(int i=1;i<=100;i++){
        if(i<2){
            isprime=0;
            continue;
        }
        isprime=1;
        t=sqrt(i);
        for(int j=2;j<=t;j++){
            if(i%j==0){
                isprime=0;
                break;
            }
        }
        if(isprime==1){
            printf("%d ",i);
        }
    }
    return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code24.c -o code24 } ; if ($?) { .\code24 }
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.25

Problem Name: Write a program that print all prime numbers, from m to n(m>n).

Source Code:

```
#include<stdio.h>
#include<math.h>
int main(){
    int m,n,isprime,t;
    printf("enter highest and lowest
number:");
    scanf("%d %d",&m,&n);


    for(int i=m;i>=n;i--){
        if(i<2){
            isprime=0;
            continue;
        }
        isprime=1;
        t=sqrt(i);
        for(int j=2;j<=t;j++){
```

```
                if(i%j==0){
                    isprime=0;
                    break;
                }
            }
            if(isprime==1){
                printf("%d",i;
            }
        }
        return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code25.c -o code25 } ; if ($?) { .\code25 }
enter highest and lowest number:20
10
19 17 13 11
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.26

Problem Name: Write a program that count total prime numbers from 1 1o n.

Source Code:

```
#include<math.h>
int main(){
    int n,isprime,t;
    printf("enter the highest number:");
    scanf("%d",&n);
    int count=0;
    for(int i=1;i<=n;i++){
        if(i<2){
            continue;
        }
        isprime=1;
        t=sqrt(i);
        for(int j=2;j<=t;j++){
            if(i%j==0){
                isprime=0;
                break;
            }
        }
        if(isprime==1){
            count=count+i;
        }
    }
    printf("the sum of prime number
is:%d",count);
    return 0;
}
```

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Output:

```
                                                                    > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code26.c -o co
de26 } ; if ($?) { .\code26 }
enter the highest number:20
the sum of prime number is:77
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.27

Problem Name: Write a program that displays first n prime numbers.

Source Code:

```
#include<stdio.h>
#include<math.h>
int main(){
    int n,isprime,count=0,t;
    printf("enter a number :");
    scanf("%d",&n);
    for(int i=1;i<=n;i++){
        if(i<2){
            continue;;
        }
        isprime=1;
        t=sqrt(i);
        for(int j=2;j<=t;j++){
            if(i%j==0){
                isprime=0;
                break;
            }
```

```
            }
            if(isprime==1){
                count++;
            }
        }
        printf("%d",count);
        return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code27.c -o code27 } ; if ($?) { .\code27 }
enter a number :20
8
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

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Problem No: 9.28

Problem Name: Write a program that print first n Fibonacci numbers.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("enter a number :");
    scanf("%d",&n);
    int a=0;
    int b=1;
    int sum;
    for(int i=1;i<=n;i++){
        sum=a+b;
        a=b;
        b=sum;
        printf("%d ",a);
    }
    return 0;
}
```

Output:

```
                                                  > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code28.c -o co
de28 } ; if ($?) { .\code28 }
enter a number :5
1 1 2 3 5
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

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Problem No: 9.29

Problem Name: Write a program that prints all Fibonacci numbers from 1 to n.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("enter a number :");
    scanf("%d",&n);
    int a=0;
    int b=1;
    int sum;
    for(int i=1;i<=n;i++){
        printf("%d ",b);
        sum=a+b;
        a=b;
        b=sum;
    }
    return 0;
}
```

Output:

```
                                                     > cd "h:\Programmi
ng\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code29.c -o co
de29 } ; if ($?) { .\code29 }
enter a number :10
1 1 2 3 5 8 13 21 34 55
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Problem No: 9.30

Problem Name: Write a program that a number Fibonacci or not Fibonacci.

Source Code:

```
#include<stdio.h>
int main(){
    int n;
    printf("enter a number :");
    scanf("%d",&n);
    int a=0;
    int b=1;
    int sum;
    for(int i=1;i<=n;i++){
        sum=a+b;
        a=b;
        b=sum;
    }
    if(a==n){
            printf("finonacci number");
    }
    else{
            printf("not a fibonacci
number");
    }
    return 0;
}
```

Output:

```
> cd "h:\Programming\Coding Practice\Basic C\exerise class work\9\" ; if ($?) { gcc code30.c -o code30 } ; if ($?) { .\code30 }
enter a number :5
finonacci number
PS H:\Programming\Coding Practice\Basic C\exerise class work\9>
```

Thank You